

Salvage Techniques For WET ELECTRONICS



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Though we all know that we should not drop electronic devices in water but wetting of electronic devices does happen accidentally. Dropping a cellphone or iPod into a pool, or spilling a glass of water or coffee on a laptop is a sure way to render the device useless. It also puts you in a jittery situation about the data stored in such devices. But not many people know what to do when this inevitably happens.

Damage due to liquids is unfortunately not covered under basic warranties, and many companies use liquid sensors to determine if a device has been exposed to water or other fluids.

Water damage is one of the more common threats to portable electronics, but sometimes if we dry these out properly, we can restore the device. This article presents some techniques that can be applied to save such electronic devices.

When liquids come in contact with circuits, currents can be inadvertently diverted (shorted) to other system components, where there can be two possible routes: burnout or grounding. When a burnout happens, a higher-current circuit (such as that from a power-handling circuit) is allowed to go through a more delicate circuit that was not designed for high currents, resulting in circuit components over-heating and melting.

When a grounding short happens, current from a circuit is merely diverted through the liquid to an alternate location that is able to handle the load, such as the computer chassis or one of many grounding circuits on the motherboard and computer components.

Shorts that directly ground a circuit have less potential for harming the circuit since grounding components are built to handle higher current loads. Liquids have a chance of doing either of these types of shorts, but often when these penetrate a device, components are just grounded instead of being burned out.

So if we are able to dry our system thoroughly, have a chance of clearing the shorts and possibly restoring functionality, especially if the liquid is relatively free of salts. But salts, even if dried, can form conductive crusts that may continue to short the circuits.

Technically, it is not water that destroys electronics, it is the small ions that are dissolved in water. These particles cause electricity to go places it is not supposed to—causing a short and ultimately causing your device to break down. Not all liquids are the same, ion-wise. Some have more ions than others. For instance, distilled water has next to no ions, whereas sea water has a ton of these.

This difference will determine exactly how much damage the liquid has caused to the electronic gadget. An electronic device wet with distilled water will probably work perfectly fine, but when it is wet with sea water, there is not much you can do to rescue it.

Unfortunately, liquids can very quickly be pulled into tight crevices and small components of circuit boards, aided by capillary action. Therefore even if a device appears to be completely dry, there may still be a small amount of liquid left over, and if it shorts even one circuit, the device may not operate properly.

Water evaporates when it comes in contact with dry air. So to dry liquids fast, it is best to have as much of it exposed to air as possible. Plugs and connections, and flat surfaces of components that are in close contact with each other, prevent the water's surface area from contacting air, and thereby retain water between these, long after the rest of the device is dry.

One way to clear this liquid out is to open the device and remove major components, and dry these off with air or absorptive materials. However, this is not feasible for many devices, given the complexity in-